INCH-POUND

MIL-C–48490B(AR)
AMENDMENT 3
30 JUNE 1998

SUPERSEDING
AMMENMENT 2
9 NOVEMBER 1987

MILITARY SPECIFICATION

CONTROL, COMMANDER'S SIGHT AND RECEIVER-TRANSMITTER, LASER RANGING, 11743121

MIL-C-48469 was inactivated after 11 July 1996 for new design.

This amendment forms a part of MIL-C–48490B (AR), dated 30 JUNE 1981, and is approved for use by the US Army Armament, Research, Development, and Engineering Center, and is available for use by all Departments and Agency of the Department of Defense.

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Table I. Loads, power, and signals:

Items 3.2 and 3.3: Delete in their entirety and substitute the following:

Item	Conditions	Characteristics	Connections
3.2	Digital type-B	Logical one: open Logical zero: 0.2 (0.2 Vdc	Applied as specified herein
3.3	Analog 0 to 36 V	0 to 36 Vrms, 400 (4 Hz, in-phase or out-of-phase with 115V, 400 Hz source of item 2.12	J2-J and J2-K
4	Auxiliary Equipment		
4.1	Electronic Supply (or equivalent)	See Appendix I	Figure I
4.2	Mirror Control Unit (or equivalent)	See Appendix II	Figure II

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3.3.1.10: Delete in its entirety and substitute the following:

"3.3.1.10 Ranging. Range conditions of the assembly shall be as follows: (See 4.6.2.10)

a. The minimum range shall be 200 meters.
b. The maximum range shall be at least 5000 meters.
c. The range accuracy shall be plus or minus 10 meters.
d. The range resolution shall be 20 meters.
e. Minimum time between ranging: When the RANGE button is depressed, the flashing red light on the RANGE button shall turn off and then turn on steady within 4 seconds.
f. The maximum number of rangings shall be 6 per minute for 2 minutes or 3 rangings per minute sustained without deterioration of range accuracy."

3.3.1.11: Delete in its entirety and substitute the following:

3.3.1.11 Minimum range inhibit. – With the optical input signal to the receiver aperture attenuated by the value calculated from the equation below (within ± 1 dB) and the logical zero type –B signal (item 3.2) of Table I applied to J2-w, the second video pulse signal as indicated in figure 6 shall not be present when the laser is fired at target 3 (Item 8a Table IX) and the RANGE readouts shall display 0.

$$\text{Attenuation value (dB)} = 40 - 20 \log (R/500)$$

Where,

R = target range in meters (actual)"

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* 3.3.1.12: Delete in its entirety.

3.3.1.13: Delete "3.3.1.13 … (See 4.6.2.13)" and substitute "3.3.1.12 … (See 4.6.2.12)".

3.3.1.14: Delete "3.3.1.14 … (See 4.6.2.14)" and substitute "3.3.1.13 … (See 4.6.2.13)".

3.3.1.15: Delete "3.3.1.15 … (See 4.6.2.15)" and substitute "3.3.1.14 … (See 4.6.2.14)".

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3.3.2.6: Delete in its entirety and substitute the following:

"3.3.2.6 Cleanliness. The optical system shall meet the cleanliness requirements of MIL-O-13830 when viewed through the eyepiece. The optical system shall show no more than five particles of dirt, 0.2 mil maximum real diameter, in the zone outside of the reticle pattern, and four particles of dirt, 0.05 mil maximum real diameter, in the zone within the reticle pattern. (See 4.6.2.21)"

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3.4: Delete in its entirety and substitute the following:

"3.4 Sealing.

3.41 Sealing. With an initial internal pressure of 4.5 ± 0.5 psig in all compartments, the pressure drop in each compartment shall not exceed a rate of 0.2 psig in a two hour period.

3.4.2 Purging. The optical and transmitter compartments shall be flushed with dry nitrogen in accordance with BB-N-411, Type I, Grade B, Class I (having a dew point at least as low as -25° F), at a pressure of 12 ± 1 psig for a 30 minute period and sealed at a pressure of 4.5 ± 0.5 psig. The electronic compartments shall be flushed at 1 psig for a five minute period, and sealed at ambient pressure."

3.5.1: Delete in its entirety and substitute the following:

"3.5.1 Shock. The assembly in its mounting configuration shall be capable of operating as specified herein after exposure to shock levels specified below. Three shock pulses shall be applied in both directions along each of the three mutually perpendicular axes (Figure 9) (See 4.6.4.1).

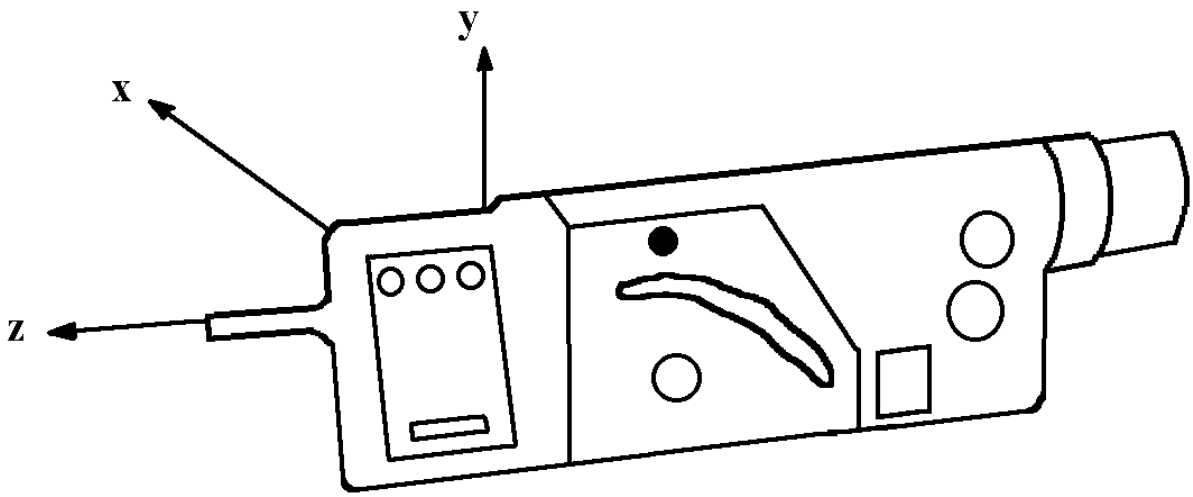


Figure 9, Test Axis

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Pulse Parameters	Basic Design	High Intensity		
		X Axis	Y Axis	Z Axis
Peak Value Duration	20g ± 3g 18 ± 3 ms	40g ± 4g * 9 ms	40g ± 4g * 9 ms	40g ± 4g * 10 ms
* Indicates saw tooth shock pulse, all others ½ sine pulse. Tolerance of saw tooth pulse in accordance with MIL-STD-810C				

3.5.1:

Line 3: Delete "40 ± 4" and substitute "20 ± 3".

Line 6: Delete "100.0 ± 10.0 g's with a time duration of 1.5 ± 0.2 msec" and substitute "40 ± 4 g's for a duration of 9 ± 1 msec".

3.5.2: Delete in its entirety and substitute the following:

"3.5.2 Vibration.

3.5.2.1 Vibration "A". The assembly in its mounting configuration shall be exposed to the vibration profile of Figure 10 for a period of 80 minutes minimum (including up to four resonance dwells of 13 ½ minutes minimum each per axis) in each of three perpendicular axes. (See 4.6.4.2.1)

3.5.2.2 Vibration "B". The assembly in its mounting configuration shall meet the requirements of 3.3.1 after being exposed to the vibration profile of Figure 10 for 15 minutes in each of three mutually perpendicular axes, 45 minutes total. (See 4.6.4.2.2)"

3.5.3: Delete in its entirety and substitute the following:

"3.5.3 Temperature – operating. The assembly shall be capable of operating as specified herein while thermally stabilized at temperatures of -25°F and +125°F. (See 4.6.4.3, 4.6.4.4, 4.6.4.5)

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Figure 9:

Title: Delete "FIGURE 9. Vibration profile." And substitute "FIGURE 10. Vibration profile.".

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3.5.4: Delete in its entirety and substitute the following:

"3.5.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure and thermal stabilization at -65(F to +160(F. (See 4.6.4.6)"

3.5.6: Sub-paragraph b, Line 2: Delete "for not less than one hour.".

Appendix I: Delete: "APPENDIX I MIRROR CONTROL UNIT REQUIREMENTS" and substitute: "APPENDIX II MIRROR CONTROL UNIT REQUIREMENTS".

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Table VI, First article tests: Delete in its entirety and substitute the following:

Item	Characteristics	Requirements	Test Procedures
1	Fabrication	3.1	Applicable drawings – Visual
2	General Specifications	3.1.2	MIL-F-13926 – Visual
3	Performance	3.3	4.6.2
4	Sealing B	3.4	4.6.3
5	Shock	3.5.1	4.6.4.1
6	Vibration A	3.5.2	4.6.4.2
7	High temperature	3.5.3	4.6.4.3
8	Low temperature	3.5.3	4.6.4.4
9	Temperature cycling	3.5.3 and	4.6.4.5
10	Temperature storage	3.5.4	4.6.4.6
11	Humidity	3.5.5	4.6.4.7
12	Altitude	3.5.6	4.6.4.8
13	Dust	3.5.7	4.6.4.9
14	Salt fog	3.5.8	4.6.4.10

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Table VII. Performance tests:

Items 112 – 131, Delete in their entirety and substitute the following:

112	A-trigger threshold and PFN low charge	3.3.1.12	4.6.2.12
113	Laser energy	3.3.1.13	4.6.2.13
114	Laser peak power	3.3.1.14	4.6.2.14
115	Sighting system test	3.3.2	4.6.2.15
116	Boresight controls – elevation	3.3.2.1	4.6.2.16
117	Boresight controls – azimuth	3.3.2.2	4.6.2.17
118	Mirror drive controls	3.3.2.3	4.6.2.18
119	Resolution	3.3.2.4	4.6.2.19
120	Parallax	3.3.2.5	4.6.2.20
121	Cleanliness	3.3.2.6	4.6.2.21
122	Image tilt	3.3.2.7	4.6.2.22
123	Parallelism of vehicle and image	3.3.2.8	4.6.2.23
124	Eyepiece focus	3.3.2.9	4.6.2.24
125	Reticle illumination	3.3.2.10	4.6.2.25
126	Operability	3.3.2.11	4.6.2.26
127	Boresight retention	3.3.2.12	4.6.2.27
128	Sight system and laser beam alignment	3.3.3	4.6.2.28
129	Indicator illumination dimmer control	3.3.4	4.6.2.29
130	Lamp test control	3.3.5	4.6.2.30
131	Sealing A	3.4	4.6.3

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Table IX, Environmental tests: Delete in its entirety and substitute the following:

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.5.1	4.6.4.1
304	Vibration B	3.5.2	4.6.4.2.2
305	High Temperature	3.5.3	4.6.4.3
306	Low Temperature	3.5.3	4.6.4.4

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307	Temperature Cycling	3.5.3	4.6.4.5
308	Temperature Storage	3.5.4	4.6.4.6
309	Sealing B	3.4	4.6.3.7

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Table X. Test Equipment:

Items 1 and 2: Delete in their entirety and substitute the following:

1	Boresight Position Holding Fixture	A rigid holding device for test assemblies having a removal / installation repeatability accuracy within 10% of any parameter tolerance.
2	Collimator	An optical device with a target reticle adjustable in azimuth and elevation (or equivalent) that shall be orientated with the boresight position holding fixture so that its target reticle can be viewed through the eyepiece of an installed assembly. The collimator target reticle center shall be aligned within 0.1 mil of the reticle center of an installed assembly having a “perfect” (i.e., no-tolerance) sighting system.

* Item 5: Delete “23-inch minimum diameter circle,” and substitute “23-inch minimum diameter circle or 23-inch square target,”.

Item 6: Delete in its entirety and substitute the following:

6	Target 3A	4 feet square minimum, 15 percent minimum diffuse reflectivity. The exact location of the target from the window of the Receiver-Transmitter shall be known within ± 1 meter.

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Item 8: Delete in its entirety and substitute the following:

8	Target 4A	4 feet square minimum, 80 percent minimum diffuse reflectivity. The exact location of the target from the window of the Receiver-Transmitter shall be known within ± 1 meter.

Item 10: Add as shown:

10	Mirror Control Unit (or equivalent)	See Appendix II

Add Figure 11 as shown:

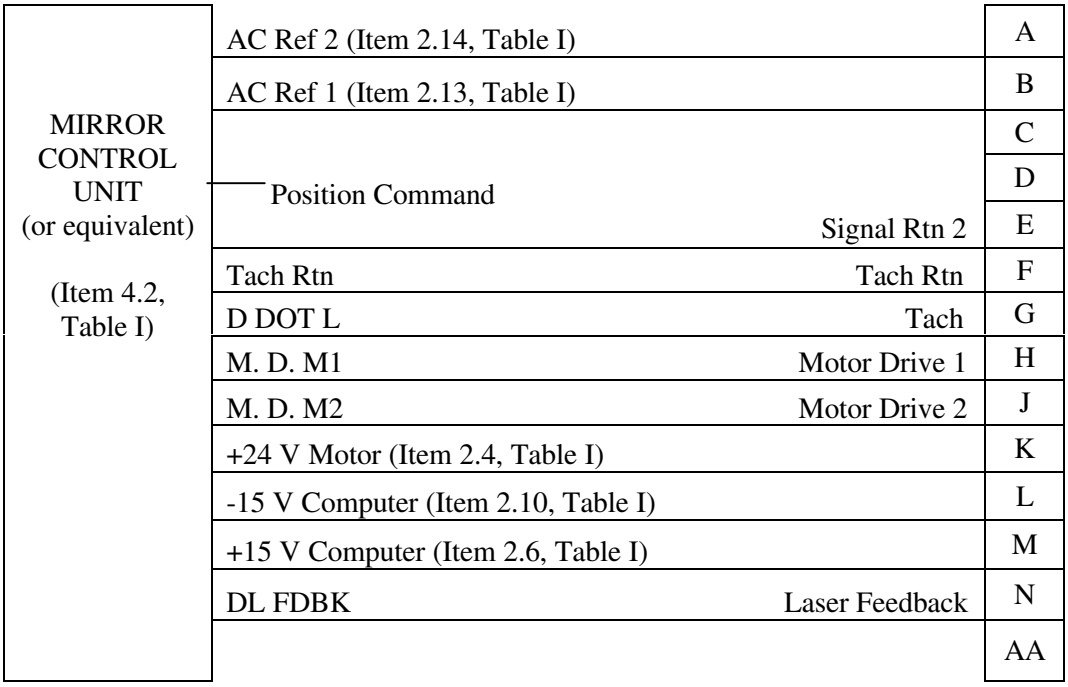


Figure 11. Receiver transmitter/mirror control interface.

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4.6.2.1: Delete in its entirety and substitute the following:

"4.6.2.1 Continuity. To determine conformance to 3.3.1.1, circuit continuity shall be measured and the resistance shall be specified in Table II with all power, signals, and loads disconnected from the assembly.

4.6.2.2: Delete in its entirety and substitute the following:

"4.6.2.2 Display readouts/lamp illumination test. To determine conformance to 3.3.1.2, apply signals to the specified pins of J-3 (Table III) and observe the display panel readouts. Apply ground to the specified pins (Table IV) and observe lamp illumination. Readouts and lamp illumination shall be as specified in Table III and Table IV."

4.6.2.3: Delete in its entirety and substitute the following:

"4.6.2.3 Thermistor resistance. To determine conformance to 3.3.1.3, the thermistor resistance measured between pins J2-u and J2-v shall be 100 ± 60 kilohms."

4.6.2.4:Delete in its entirety and substitute the following:

"4.6.2.4 Lasing pulse timing. To determine conformance to 3.3.1.4, the time elapsed between the start of SCR trigger and start of A-trigger, measured at pin J2-P and J2-n, respectively, shall conform to Figure 1."

4.6.2.5: Delete in its entirety and substitute the following:

"4.6.2.5 A-trigger waveforms. To determine conformance to 3.3.1.5, the unit shall be fired and waveforms measured J2-m and J2-n. Waveforms shall be shown in Figure 2."

4.6.2.6: Delete in its entirety and substitute the following:

"4.6.2.6 Malfunction 3 signal. To determine conformance to 3.3.1.6, the voltage shall be measured at J2-f and shall be as follows:

a. Fire the units and observe the output at J2-f and verify that the voltage is 4.2 ± 1.2V.

b. Remove the voltage from J2-s. Apply the logical zero type-A signal of Table I to J2-R, and observe the voltage at J2-f. Verify that the voltage is 0.2 ± 0.2V.

c. Momentarily apply the logical zero type-A signal of Table I to J2-e. Verify that the voltage is 4.2 ± 1.2V."

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4.6.2.7: Delete in its entirety and substitute the following:

"4.6.2.7 Range and video signal. To determine conformance to 3.3.1.7, apply the digital type-A signals specified to J2-h and J2-I (Figure 3) and verify that the video signal output waveforms are as shown in Figure 4."

4.6.2.8: Delete in its entirety and substitute the following:

"4.2.6.8 Reset signal. To determine conformance to 3.3.1.8, the assembly shall be fired and the output at J2-c and J2-d observed. The waveforms shall be as shown in Figure 5."

4.6.2.9: Delete in its entirety and substitute the following:

"4.2.6.9 ACC inhibit. To determine conformance to 3.3.1.9, cover the receiver aperture, apply the digital type-A signal to J2-h and J2-I and observe the ACC inhibit between J2-b and J2-V (Rtn). The output shall be 29.0 ± 4.0 V. Remove the digital type-A signals and again observe the output. Verify the voltage is 0.0 ± 1.0 V."

4.6.2.10: Delete in its entirety and substitute the following:

"4.6.2.10 Ranging.

4.6.2.10.1 Minimum range test. To determine conformance to 3.3.1.10.1, perform the following:

a. Perform 5 ranging operations against Target 1 (Item 4, Table X).

b. Observe range readouts. Readouts shall be actual target range within the specified range accuracy (see 3.3.1.10.c).

4.6.2.10.2 Maximum range test. Maximum ranging will be simulated by an extinction test by placing neutral density filters over the receiver aperture. To verify conformance to 3.3.1.10.b, perform the following:

a. Connect J2-q to J2-V and place neutral density filters over the receiver aperture with an attenuation of 40 ± 1 db for target No. 4A or 36 ± 1 db for target No. 4B.

b. Perform 10 ranging operations against Targets 4A or 4B (Items 8 and 9, Table 10). The target shall be first, normal to the laser beam within ± 10 degrees, with a minimum reflectivity of 80 percent.

c. Observe range readouts. At least 9 of 10 readouts shall be the actual target range within the specified range accuracy (see 3.3.1.10.c).

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d. Disconnect J2-q from J2-V and place neutral density filters over the receiver aperture with an attenuation of 10.0 ± 0.5 db for target No. 4A or 6.0 ± 0.5 db for target No. 4B.

e. Perform five ranging operations against target No. 4A or 4B.

f. Observe range readouts. The readouts shall be the actual target range within the specified range accuracy.

4.6.2.10.3 Range accuracy. To determine conformance to 3.3.1.10.c, perform the following:

a. Perform 5 ranging operations against Target 3A or #B (Items 6 and 7, Table X).

b. Observe range readouts. Verify the accuracy of the indicated range.

4.6.2.10.4 Range resolution. To determine conformance to 3.3.1.10.d, perform the following:

a. Perform 5 ranging operations against target 2 and 3A or 3B (Items 5, 6, and 7, Table X). Targets shall be aligned so as to allow the transmitted laser beam to strike both targets.

b. Observe the readouts. Readouts shall be the actual range to each of the targets within the specified range accuracy (see 3.3.1.10.c)."

4.6.2.11: Delete in its entirety and substitute the following:

"4.6.2.11 Minimum range inhibit.

4.6.2.11.1 First article and special sampling. To determine conformance to 3.3.1.11, attenuate the input signal to the receiver aperture 40 ± 1 db for target No. 3A or 36 ± 1 db for target No. 3B. Apply the logical zero type-B signal to J2-w. Fire the unit at the target defined in Table X, items 6 or 7, and observe the output at J2-j. Verify that the second video pulse signal as shown in figure 7 is not present.

4.6.2.11.2 Performance. To determine conformance to 3.3.1.11, attenuate the input signal to the receiver aperture 40 ± 1 db for target No. 3A or 36 ± 1 db for target No. 3B. Apply the logical zero type-B signal to J2-w. Fire the unit at the target defined in Table X, items 6 or 7, and observe the RANGE and RETURNS readout displays 0."

* 4.6.2.12: Delete in its entirety.

4.6.2.13: Delete in its entirety and substitute the following:

"4.6.2.12 A-trigger threshold and PFN low charge. To determine conformance to 3.3.1.12, set the energy output of the assembly to 15 ± 3 millijoules. Fire the unit and observe the voltage at J2-f. The voltage shall be 4.2 ± 1.2 Vdc. Apply 10.0 ± 1.0 Vdc to J2-x and fire the unit again. The measured voltage shall be 0.2 ± 0.2 Vdc."

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4.6.2.14: Delete in its entirety and substitute the following:

"4.6.2.13 Laser energy test. To determine conformance to 3.3.1.13, perform the following:

a. Prior to the operation, adjust R10 of the PFN Charge Power Supply, or its equivalent, for an energy output of 50 ± 5 millijoules.

b. Fire the laser and record the energy and PFN current.

The PFN current shall be less than the value indicated in Figure 8 for the corresponding thermistor resistance.

NOTE: R10 shall not be further adjusted from this initial setting before, during, or after the environment tests."

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4.6.2.15: Delete in its entirety and substitute the following:

"4.6.2.14 Laser peak power test. To determine conformance to 3.3.1.14, measure the peak output power by one of the following methods:

a. Measure the laser output energy and the laser pulse duration at the one-half peak-power points. The ratio of energy to pulse duration shall be at least one megawatt, or

b. Measure the peak-power directly from an oscilloscope display of the output of a calibrated high speed photodiode. The peak power shall be at least one megawatt."

Add paragraph 4.6.2.15:

"4.6.2.15 Sighting systems test. To establish the boresight position specified in 3.3.2 for each assembly under test, install the assembly on the boresight position holding fixture with collimator (Items 1 and 2, Table X) and perform the following:

a. Rotate the elevation boresight knob of the assembly to align the reticle with the target image in elevation.

b. Connect the assembly to the mirror control unit (Item 10, Table X) as shown on Figure 11, close the servo loop, and adjust the azimuth position control on the mirror control unit to align the reticle with the target image in azimuth.

NOTE: Readjust the elevation boresight knob and the azimuth position control, if necessary, to maintain alignment."

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4.6.2.16: Delete in its entirety and substitute the following:

"4.6.2.16 Boresight controls – elevation.

4.6.2.16.1 Boresight knob travel. To determine conformance to 3.3.2.1.1, the reticle shall be viewed through the eyepiece as the elevation boresight knob is rotated. Verify that the reticle moves a minimum of plus or minus 5.0 mils in relation to the collimator reticle and that clockwise rotation of the boresight knob moves the reticle upwards.

4.6.2.16.2 Boresight knob accuracy. To determine conformance to 3.3.2.1.2, boresight knob accuracy shall be measured by rotating the knob through a given number of graduations on the boresight knob scale as the assembly reticle movement is compared to the graduations of the collimator reticle. Measurements shall be made at plus and minus 2.0 and 4.0 mils. Graduation accuracy shall be as specified.

4.6.2.16.3 Backlash. To determine conformance to 3.3.2.1.3, backlash shall be measured in accordance with the following procedures:

a. Rotate the elevation boresight knob in a cw direction to a given position on the boresight knob scale.

b. Observe and record the elevation position of the assembly reticle center as superimposed on the collimator target image.

c. Continue rotation of the boresight knob several mils in a cw direction.

d. Rotate the boresight knob in a ccw direction to the position established in step "a".

e. Observe and record the assembly reticle elevation position as in step "b". The difference between this reading and that recorded in step "b" is the backlash.

Backlash in accordance with steps a through e shall be measured at approximate boresight position and plus and minus 4.0 mils elevation. Backlash shall not exceed 0.1 mil.

4.6.2.16.4 Reticle shift. To determine conformance to 3.3.2.1.4, the boresight knob shall be rotated through the range specified as the assembly reticle shift is measured in relation to the collimator reticle. Reticle shift shall not exceed 0.1 mil."

4.6.2.17: Delete in its entirety and substitute the following:

"4.6.2.17 Boresight controls – azimuth. 5.000 ± 0.002 Vrms shall be applied (AC Ref 1 and AC Ref 2) as specified in Item 2.13 and 2.14 of Table I and tests shall be conducted as specified in 4.6.2.17.1 through 4.6.2.17.3.

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4.6.2.17.1 Null position. To determine conformance to 3.3.2.2.1, the mirror shall be set to the boresight position and the azimuth boresight knob shall be adjusted to achieve a minimum voltage output of 10 mVrms or less, measured at J2-AA.

4.6.2.17.2 Boresight knob travel. To determine conformance to 3.3.2.2.2, the feedback voltage shall be measured at J2-AA, from the null position established in 4.6.2.17.1, as the boresight knob is rotated to two directional extremes. Minimum voltage output shall be 295 mVrms at each extreme of travel. Verify that the feedback voltage is in phase with AC Ref 1 (J2-B) at the extreme clockwise position.

4.6.2.17.3 Boresight knob accuracy. To determine conformance with 3.3.2.2.3, boresight knob accuracy tests shall be conducted at the null position established in 4.6.2.17.1. The boresight knob shall be rotated 3.5 mils and 5.0 mils clockwise and counterclockwise per the boresight knob scale, as feedback voltage is monitored at J2-AA. The measured voltage shall be 206 ± 9 mVrms at 3.5 and 295 ± 9 mVrms at 5.0 mils."

4.6.2.18: Delete in its entirety and substitute the following:

"4.6.2.18 Mirror drive controls.

4.6.2.18.1 Mirror (resolver) accuracy. To determine conformance to 3.3.2.3.1, tests for mirror drive accuracy shall be conducted at the mirror position specified in Table V. With the assembly boresighted on the collimator described in 4.6.2.15 and the null position established in 4.6.2.17.1, an alternating voltage shall be applied to the mirror drive motor to position the mirror at each target position specified in Table IV. The laser feedback voltage shall be as specified.

4.6.2.18.2 Reticle shift (elevation). To determine conformance to 3.3.2.3.2, the mirror drive motor shall be energized to rotate the mirror through a 36 mils left and right travel, from boresight position, as the assembly reticle is compare to the horizontal line of the collimator reticle, defined in 4.6.2.15. In tests conducted, the assembly reticle shall not deviate more than 0.15 mils above or below the horizontal line of the collimator reticle, starting from a coincidence condition at the boresight position.

4.6.2.18.3 Mirror slew rate. To determine conformance to 3.3.2.3.3, the mirror drive motor shall be energized with 56.0 ± 1.0 Vrms and the time required for 10 complete cycles of mirror travel shall be measured. Time required in each case shall not exceed 22.0 seconds at standard ambient temperature and 40.0 seconds at temperature extremes."

4.6.2.18.4 Tachometer outputs. To determine conformance to 3.3.2.3.4, the following tests shall be conducted:

a. With the azimuth boresight knob in the null position as defined in 4.6.2.17.1, the tachometer output voltage shall be measured between J2-H and J2-G return.

b. With 36.0 ± 0.1 Vrms at 400 Hz applied first at J2-J and then at J2-K to drive the mirror motor, the tachometer output voltage shall be measured between J2-H and J2-G return and shall be a minimum of 2.5 Vrms.

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4.6.2.18.5 Mirror-servo stability. To determine conformance to 3.3.2.3.5, connect the unit to the test circuit as specified in Figure 11. Measure the mirror oscillation optically to verify that the requirements of 3.3.2.3.5 are met.

4.6.2.18.6 Mirror-servo system overshoot. To determine conformance to 3.3.2.3.6, with the assembly operating as specified, the mirror shall be made to slew to a position requiring approximately 25 mils of mirror travel as the feedback voltage at J2-AA is monitored for amount and duration of overshoot. The monitored voltage shall not exceed 0.5 Vrms and shall indicate that the mirror has achieved a stable condition at the new position within a period of 250 milliseconds."

4.6.2.19: Delete in its entirety and substitute the following:

"4.6.2.19 Resolution. To determine conformance to 3.3.2.4, resolution tests shall be conducted in accordance with MIL-O-13830, except that the USAF 1951 resolving power chart may be used. Resolving power shall be as specified in the requirements of 3.3.2.4."

4.6.2.20: Delete in its entirety and substitute the following:

"4.6.2.20 Parallax. To determine conformance to 3.3.2.5, parallax shall be tested when viewing a target, or a simulated target, located at a distance of 1000 ± 100 meters. There shall be no apparent movement greater than 0.10 mils between the reticle and the target image as the observer moves his eyes from the center of field of view to either horizontal or vertical field of view edge."

4.6.2.21: Delete in its entirety and substitute the following:

"4.6.2.21 Cleanliness. The optical surface shall be inspected for cleanliness in accordance with MIL-O-13830. The optical system shall conform to the cleanliness requirements of 3.3.2.6."

4.6.2.22: Delete in its entirety and substitute the following:

"4.6.2.22 Image tilt. To determine conformance to 3.3.2.7, an auxiliary telescope shall be used in conjunction with the collimator to check image tilt of the assembly. See 4.6.2.15. With the auxiliary telescope originally oriented so that its reticle is parallel to the collimator reticle (without the assembly in its holding fixture), any tilt or rotation of the image shall then be determined by viewing the collimator reticle through the assembly and measuring the parallel deviations of the auxiliary telescope reticle to the collimator. The two reticles shall be parallel within two degrees."

4.6.2.23: Delete in its entirety and substitute the following:

"4.6.2.23 Parallelism of reticle and image. To determine conformance to 3.3.2.8, the image of the collimator image shall be compared for parallelism of the assembly while viewing through the assembly eyepiece. The two reticles shall be parallel within one degree."

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4.6.2.24: Delete in its entirety and substitute the following:

"4.6.2.24 Eyepiece focus. To determine conformance with 3.3.2.9, the focus shall be tested with the aid of a calibrated dioptometer having a magnification of at least three power (3X), and a diopter measuring range of at least plus and minus 4 diopters. Set the dioptometer diopter scale to zero, and adjust the dioptometer eyepiece for the best focus on the dioptometer reticle. Position the dioptometer at the telescope eyepiece. Obtain best focus of the telescope reticle at the center of the field by adjustment of the dioptometer diopter scale. The assembly reticle shall be in best focus within the tolerance specified in 3.3.2.9, as measured on the dioptometer diopter scale. Set the diopter scale of the assembly to its extreme plus setting, and obtain the best focus of the assembly reticle at the center of the field by adjustment of the dioptometer diopter scale. The focus of the assembly shall be no less than+280 diopters as measured with the dioptometer. Repeat this test with the diopter scale of the scale of the telescope set to its extreme minus setting. Eyepiece focus of the assembly shall conform to3.3.2.9. The test shall be conducted at 6X and 12X."

4.6.2.25: Delete in its entirety and substitute the following:

"4.6.2.25 Reticle illumination. To determine conformance to 3.3.2.10, the reticle light control knob shall be rotated and the reticle viewed for assurance of increased brightness."

4.6.2.26: Delete in its entirety and substitute the following:

"4.6.2.26 Operability.

4.6.2.26.1 Diopter adjustment torque. To determine conformance to 3.3.2.11.1, a torque measuring device shall be attached to the eyepiece and the torque shall be measured as the eyepiece is rotated throughout its full travel, clockwise and counterclockwise. The torque required shall be as specified.

4.6.2.26.2 Boresight knob torque. To determine conformance to 3.3.2.11.2, a torque measuring device shall be used to measure the torque required to rotate the boresight knobs throughout their full travel in both directions. The torque required shall be as specified.

4.6.2.26.3 Boresight knob disengagement. To determine conformance to 3.3.2.11.3, a force measuring device shall be attached to the boresight knobs and the force required to disengage the back knob shall be measured. The force required shall be as specified.

4.6.2.26.4 Magnification knob torque. To determine conformance to 3.3.2.11.4, a torque measuring device shall be attached to the magnification change knob and the torque to change from 6X to 12X and back to 6X shall be measured. The torque required shall be as specified."

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4.6.2.27: Delete in its entirety and substitute the following:

"4.6.2.27 Boresight retention. To determine conformance to 3.3.2.12, tests shall be conducted in accordance with the following:

a. Establish boresight position by aligning the assembly reticle on the collimator reticle described in 4.6.2.15.

b. Establish the null position as described in 4.6.2.17.1.

c. Slip the boresight knob scales to 4.0 and provide assurance that the scales do not move for the remainder of the test.

d. Subject the unit to environmental tests required.

e. Return the unit to its boresight fixtures.

f. Apply voltage, if necessary, to the motor drive to obtain a null (<10 mVrms) at J2-AA.

g. The change in boresight position from that established in step a, as indicated by the deflection of the assembly reticle position as superimposed on the collimator target image, shall not exceed the amount specified in 3.3.2.12."

4.6.2.28: Delete in its entirety and substitute the following:

"4.6.2.28 Sighting system/laser beam alignment.

4.6.2.28.1 Initial alignment. To determine conformance to 3.3.3.1, mount a long focal length lens in front of the assembly window and place a piece of thermally sensitive paper containing target grids in the focal place of the lens. Align the reticle on a target grid juncture and fire the assembly. Repeat this procedure five times. Examine the burn spots under a microscope and determine the average deviation of the burn spots from the grid juncture. The average deviation shall not exceed 0.1 mil.

4.6.2.28.2 Alignment retention. To determine conformance to 3.3.3.2, the exact alignment established in 4.6.2.28.1 shall be recorded, the assembly shall be subjected to environmental tests or run-in tests, as required, and the alignment shall again be checked. The change in alignment shall not exceed that specified in 3.3.3.2."

4.6.2.29: Delete in its entirety and substitute the following:

"4.6.2.29 Indicator illumination dimmer control test. To determine conformance to 3.3.4, rotate the dimmer control knob in the clockwise direction up to but not into the first TEST position. Verify that the FEED and RESET indicator lamps change from dim to bright. Rotate the lamp in the counterclockwise direction to the DIM position. Verify that the FEED and RESET indicator lamps change from bright to dim."

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4.6.2.30: Delete in its entirety and substitute the following:

"4.6.2.30 Lamp test control. To determine conformance to 3.3.5, apply power and digital type-A signals to the assembly as specified therein. Rotate the dimmer control fully clockwise into the TEST position. Measure the voltage between J3-r (+) and J3-M (rtn). The voltage shall be 0.2 ± 0.2 Vdc. Verify that the thousands readout displays the digit 8. Rotate the dimmer control counterclockwise out of the TEST position. Measure the voltage between J3-r (+) and J3-M (rtn). The voltage shall be 4.2 ± 1.2 V. Verify that the readout displays the digit 3."

4.6.3: Delete in its entirety and substitute the following:

"4.6.3 Sealing. Flush each compartment as specified in 3.4.2. Pressurize each compartment with dry nitrogen (having a dew point at least as low as -25°C) at 4.5 ± 0.5 psig and subject the assembly to one of the following leakage tests:

NOTE: In the event of conflict, Sealing B shall take precedence over Sealing A.

NOTE

After successfully completing test (a) or (b), pressurize the optical and transmitter compartments with 4.5 ± 0.5 psig of dry nitrogen and reseal; reduce the pressure in the compartment behind the commander's panel to ambient and reseal."

4.6.3.1: Delete in its entirety and substitute the following:

"4.6.3.1 Sealing A. Use item 3 of table X and subject the assembly to the leakage (immersion) test specified in MIL-STD-810, method 512.1, procedure IV. Immerse the assembly for a period of 5 minutes in a suitable liquid such as water so that the uppermost part of the equipment is below the surface of the liquid. There shall be no leakage as evidence by bubbles coming from within the equipment. Bubbles that result from trapped air on the various surfaces of the test item shall not be considered leakage."

4.6.4: Delete in its entirety and substitute the following:

"4.6.4 Environmental tests. The storage temperature tests of 4.6.4.5 shall be performed after all other environmental tests have been performed.

a. Assemblies selected to undergo the qualification inspections of Table VI shall be subjected to the performance tests as follows:

1. Pre-environmental tests. Subject each assembly to all the performance tests of 4.6.2 prior to the performance of each environmental test of 4.6.4, except 4.6.2.28.

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2. During environmental tests. Subject each assembly to the selected performance tests of 4.6.2 only when specified in an environmental test of 4.6.4.

3. Post-environmental tests. Unless otherwise specified, subject each assembly to all the performance tests of 4.6.2 after the performance of each environmental test of 4.6.4, except 4.6.2.28 and in test 4.6.2.16 the measurement shall be made at plus and minus 2.0 mils.

b. Each assembly undergoing conformance inspections of Table IX shall have passed the performance tests in Table VII as indicated therein. Where specified by the control test method, the assembly shall be subjected to performance tests during and immediately after each environmental test performed. After all environmental tests have been completed, the assembly shall meet the performance tests of 4.6.2 except 4.6.2.28 and in test 4.6.2.16 the measurement shall be made at plus and minus 2.0 mils."

4.6.4.1.1: Delete in its entirety and substitute the following:

"4.6.4.1.1 Basic design. Mount the assembly in shock Test Fixture 11738146 (or equivalent). With power applied, subject the assembly to the test specified in MIL-STD-810C, method 516.2, procedure I shall be used. Apply three shock pulses in both directions along the three mutually perpendicular axes. Shock pulse configuration, peak amplitude, and time duration measured at the 10 percent amplitude points shall be in accordance with 3.3.3.6. During this test, the assembly shall be monitored to verify that there is no change in the range or return readouts. After the test (18 shocks) the assembly shall be examined for evidence of damage and subjected to the tests of 4.6.2.2, 4.6.2.4, 4.6.2.5, 4.6.2.9, 4.6.2.10, 4.6.2.15, 4.6.2.18.3, 4.6.2.18.4, 4.6.2.27, and 4.6.2.28.2 to determine compliance with 3.5.1."

4.6.4.1.2: Delete in its entirety and substitute the following:

"4.6.4.1.2 High intensity. This test shall be the same as 4.6.4.1.1 except MIL-STD-810C, method 516.2, procedure IV shall be used."

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4.6.4.2.1: Delete in its entirety and substitute the following:

"4.6.4.2.1 Vibration "A". Subject the assembly in its mounting configuration to the vibration test specified in MIL-STD-810C, Method 514.2, Procedure VII, except as specified herein. Test conditions shall be in accordance with 3.5.2.1 and the following:

a. Connections and instrumentation shall be attached to the assembly to permit operation and testing as specified in b and c.

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b. Conduct resonance frequency search along each axis at room temperature at reduced vibration amplitude, without input power, and over the frequency range of 5 to 500 Hz. Up to four resonant frequencies shall be identified (the four most severe resonant frequencies if more than four are present) in each axis for the resonance dwell test. A resonance is defined as an amplification greater than three times the input amplitude.

c. During the resonance dwell periods (13-1/3 minutes at each resonant frequency) and the sinusoidal cycling tests, the input power shall be applied and the assembly shall be subjected to and pass the performance tests of 4.6.2.2, 4.6.2.7, 4.6.2.18.3, and 4.6.2.18.4.

d. After vibration, the assembly shall be examined for damage and shall pass the performance tests of 4.6.2 except 4.6.2.28.1 to determine compliance with 3.5.2.1."

4.6.4.2.2: Delete in its entirety and substitute the following:

"4.6.4.2.2 Vibration "B". This test shall be conducted as in 4.6.4.2.1, except as follows:

a. Resonance search and dwell shall not be conducted.

b. The period of vibration shall be 15 minutes in each axis, 45 minutes total.

c. After vibration, the assembly shall be examined for damage and shall pass the performance tests of 4.6.2, except 4.6.2.28.1, to determine compliance with 3.5.2.2."

4.6.4.3: Delete in its entirety and substitute the following:

"4.6.4.3 High temperature. Subject the assembly to the following high temperature test to determine compliance with 3.5.3.

a. Install the assembly in the test chamber.

b. With the assembly not operating, increase the initial test chamber temperature to 125°F at ambient pressure at a rate of 3°F per minute or less.

c. Allow the assembly temperature to stabilize for a period of five hours.

d. With the assembly temperature stabilized, and the internal test chamber at 125°F, subject the assembly to the test performance tests specified by 4.6.2.2, 4.6.2.5, 4.6.2.9, 4.6.2.10.3, 4.6.2.15, 4.6.2.18.3, and 4.6.2.18.4.

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e. With the assembly nonoperating, reduce the internal test chamber temperature to an ambient temperature of 73 ± 18°F at a rate of 3°F per minute or less.

f. Allow the assembly temperature to stabilized at ambient temperature for a period of five hours and subject the assembly to the performance tests of 4.6.2, except 4.6.2.28.1."

4.6.4.4 Delete in its entirety and substitute the following:

"4.6.4.4 Low temperature. Subject the assembly to the following low temperature test to determine compliance with 3.5.3.

a. Install the assembly in the test chamber.

b. With the assembly nonoperating, reduce the internal test chamber temperature to -25° at ambient pressure at a rate of 3°F per minute or less.

c. With the assembly temperature stabilized for a period of five hours, and the internal test chamber temperature at -25°F, subject the assembly to the performance tests specified by 4.6.2.2, 4.6.2.5, 4.6.2.9, 4.6.2.10.3, 4.6.2.15, 4.6.2.18.3, and 4.6.2.18.4.

d. With the assembly nonoperating, increase the internal test chamber to an ambient temperature of 73 ± 18°F at a rate of 3°F per minute or less.

e. Allow the assembly temperature to stabilize at ambient temperature for a period of five hours and subject the assembly to the performance tests specified by 4.6.2 and 4.6.2.27."

4.6.4.6: Delete in its entirety and substitute the following:

"4.6.4.6 Temperature-storage. Subject the assembly to the following temperature-storage test to determine compliance with 3.5.4.

a. Install the assembly in the test chamber.

b. With the assembly nonoperating, reduce the internal test chamber temperature to -65°F at the rate of 3°F per minute or less and maintain this temperature for a minimum period of five hours.

c. Increase the internal test chamber temperature to 160°F at a rate of 3°F per minute or less and maintain this temperature for a minimum period of five hours.

d. Reduce the internal test chamber temperature to ambient conditions (+75°F ± 18°F) at a rate of 3°F per minute or less.

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4.6.4.7: Delete in its entirety and substitute the following:

"4.6.4.7 Humidity. Place the assembly in a humidity chamber and subject it to the humidity test specified in MIL-STD-810C, method 507.1 procedure IV. During humidity exposure, verify that the assembly passes the performance tests of 4.6.2.2, 4.6.2.5, 4.6.2.9, 4.6.2.18.3, and 4.6.2.18.4. After exposure, verify that the assembly passes all the performance tests of 4.6.2 except 4.6.2.27 and 4.6.2.28.1."

4.6.4.8: Delete in its entirety and substitute the following:

"4.8.4.8 Altitude. Place the assembly in an altitude chamber and subject it to the altitude test specified in MIL-STD-810C, method 500.1, procedure I, except that:

a. The assembly, in nonoperating condition shall be exposed to an atmospheric pressure equivalent to 40,00 feet above sea level for a period of not less than one hour.

b. With the chamber pressure increased to 10,00 feet above sea level, the assembly shall be subjected to the performance tests of 4.6.2.2, 4.6.2.4, 4.6.2.5, 4.6.2.9, 4.6.2.18.3, and 4.6.2.18.4.

After exposure subject the assembly to the performance tests of 4.6.2 except 4.6.2.27 and 4.6.2.28.1."

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4.6.4.9: Delete in its entirety and substitute the following:

"4.6.4.9 Dust. Subject the assembly to the dust test specified in MIL-STD-810, method 510.1, procedure I. At the conclusion of this test, verify that the assembly passes the performance tests of 4.6.2 except 4.6.2.27 and 4.6.2.28.1."

Add paragraph 4.6.4.10:

"4.6.4.10 Salt fog. Subject the assembly to the corrosion test specified in MIL-STD-810C, method 509.1, procedure I. At the conclusion of this test, verify that the assembly passes the performance tests of 4.6.2 except 4.6.2.28.1 and 4.6.2.27."

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Figure 10: Delete "Figure 10. Temperature cycling" and substitute "Figure 12. Temperature cycling".

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Add Appendix 1 following paragraph 6.3 as shown:

"APPENDIX I

ELECTRONIC SUPPLY REQUIREMENTS

10.1 SCOPE

10.1.1 This specification establishes the performance requirements for the Electronic Supply Special Test Equipment.

10.2 APPLICABLE DOCUMENTS

11743131	Electronic Unit
11743372	Circuit Card Assembly
11743142	Control, Commanders, Laser Ranging, A30A1, Schematic Design
MIL-C-48490B (ARRADCOM)	Military, Specification, Control, Commander's Sight and Receiver-Transmitter, Laser Ranging: 11743121

10.3 REQUIREMENTS

10.3.1 Performance. The Electronic Supply Special Test Equipment shall meet the requirements of this specification under the following conditions:

a.	Temperature	73 ± 18°F
b.	Pressure	28.5 +2.0 -4.5 inches of mercury
c.	Relative Humidity	50 ± 30%
d.	Power, Signals, Loads and Interconnections	See Table I

10.3.1.1 Initial turn on. With a logical zero (Item 3.1, Table I) applied to the range switch 1 input, and with the application of power (Item 2.0 Table I), the following conditions shall exist:

a. The voltage waveform as specified on Figure 2 shall be present between the range light return output and the return line.

b. The voltage between the last light return output and the return line shall be 0.4 ± 0.3 Vdc.

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TABLE I. Loads, power, and signals.

Item	Conditions	Characteristics	Connections
1.0	Loads		Connect items 1.1 through 1.11 between the points indicated
1.1	Fixed resistor	270 ohms ± 5 percent, 2 watts	go light return and 14 Vdc (Item 2.4) test light return and 14 Vdc (Item 2.4) ~~on light return~~ and 14 Vdc (Item 2.4) select light return and 14 Vdc (Item 2.4)
1.2	Fixed resistor	180 ohms ± 5 percent, 2 watts	range light return and 14 Vdc (Item 2.4) first light return and 14 Vdc (Item 2.4) second light return and 14 Vdc (Item 2.4) last light return and 14 Vdc (item 2.4)
1.3	Fixed resistor	100 kilohms ± 1 percent, ½ watt	Unless otherwise specified, Thermistor and Thermistor return.
1.4	Fixed resistor	32.4 kilohms ± 1 percent ½ watt	Thermistor and Thermistor return when specified
1.5	Fixed resistor	2 kilohms ± 5 percent ¼ watt	When specified
1.6	Fixed resistor	1 kilohms ± 5 percent ¼ watt	Fire and 5 Vdc (Item 2.2)
Note: Reference Figure 1			

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TABLE 1. Loads, power, and signals. – Continued.

Item	Conditions	Characteristics	Connections
1.7	Fixed resistor	470 ohms ± 5 percent ½ watt	Anode 1 and return Anode 2 and return Anode 3 and return Anode 4 and return Anode 5 and return
1.8	Fixed resistor	1.5 kilohms, ± 5 percent, ¼ percent	Data 1 and 5 Vdc (Item 2.2) Data 2 and 5 Vdc (Item 2.2) Data 4 and 5 Vdc (Item 2.2) Data 8 and 5 Vdc (Item 2.2) manual reset and 5 Vdc (Item 2.2)
1.9	Fixed resistor	330 ohms, ± 5 percent 1 watt	minimum range inhibit ≥ and 5 Vdc (Item 2.2)
1.10	Fixed resistor	1.8 kilohms, ± 5 percent, ¼ watt	blanking and 5 Vdc (Item 2.2)
1.11	Fixed resistor	180 ohms, ± 5 percent 2 watts	RBI and 5 Vdc (Item 2.2)
2.0	Power Source		
2.1	24 Vdc	24 ± 2.0 Vdc Ripple: ≤ 200 mV p-p	
2.2	5 Vdc	Tolerance: ± 0.2 Vdc Ripple: ≤ 25 mV p-p	
2.3	15 Vdc	Tolerance: ± 0.7 Vdc Ripple: ≤ 25 mV p-p	
2.4	14 Vdc	Tolerance: ± 0.5 Vdc Ripple: ≤ 25 mV p-p	
2.5	29 Vdc	Tolerance: ± 1 Vdc Ripple: ≤ 20 mV p-p	

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TABLE 1. Loads, power, and signals. – Continued.

Item	Conditions	Characteristics	Connections
2.6	-5 Vdc	Tolerance: ± 0.2 Vdc Ripple: ≤ 25 mV p-p	
2.7	24 'Q' SW	24.0 ± 2.0 Vdc Ripple: ≤ 200 mV	
2.8	DC return		Figure I
3.0	Signals		
3.1	Digital type-A	Logical one: 4.0 ± 1.0 Vdc Source independence: 2 k ohms, ± 5 percent Logical zero: 0.2 ± 0.2 Vdc	As specified herein
4.0	Auxiliary equipment		
4.1	Readouts	Circuit Card Assembly 11743372 or equivalent	Figure 5 when specified

10.3.1.2 PFN current sensing. With the PFN voltage set at 1200 ± 15 V between the +1600 V coil return line, the current between the PFN voltage sense output and the dc return line shall be 75 ± 2.5 microamperes.

10.3.1.3 PFN voltage adjustment/programming.

10.3.1.3.1 PFN voltage adjustment. The PFN voltage between the +1600 V coil and the +1600 V coil return shall be adjustable over a range of 1050 Vdc to 1300 Vdc with a 100k ohm load (Item 1.3, Table I) between the thermistor and the thermistor return lines.

10.3.1.3.2 PFN voltage programming. With the PFN voltage between the +1600 V coil output and the +1600 V coil return set at 1200 ± 15 V, changing the load between the thermistor and the thermistor return lines from 100 kilohms (Item 1.3, Table I) to 32.4 kilohms (Item 1.4, Table I) shall cause the PFN voltage to increase to 1345 ± 40 V.

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10.3.1.4 400 V supply. With the PFN voltage between the +1600 V coil output and the +1600 V coil return set at 1200 ± 200 V, the voltage between the +400 V and +400 V return lines shall be 400 ± 40 V.

10.3.1.5. Fire. With the PFN voltage between the +1600 V coil output and the +1600 V coil return set at 1200 ± 200 Vdc, with the input signal conditions established as listed in Table II for each item listed therein, and with the Table II inputs established in the order of items 1 through 5, the corresponding order of output signals shall be as specified therein.

10.3.1.6 Range readout. With the range counters removed from the electronic supply and with the input signal conditions established as specified in Table III, The output signal conditions shall be as specified in Table III.

10.3.1.7 Manual reset. With a logical zero (Item 3.1, Table I) applied between the manual reset switch and the and the dc input return line the voltage between the manual reset and return lines shall be 0.2 ± 0.2 Vdc. With the manual reset switch input open circuited, the voltage between the manual reset and return lines shall be 4.5 ± 1.0 Vdc.

10.3.1.8 Switch and light control. With the input signal conditions of Table IV established in any item order, the output signal conditions shall be as specified in Table IV.

10.3.1.9 Blanking. With the momentary (≥ 1 second) application of a logical zero (Item 3.1, Table I) in sequence between:

a. manual reset switch and dc return,

b. range switch 1 and dc return, and

c. range switch 2 and dc return,

the waveform of Figure 4 shall appear upon application of step C at the blanking output.

10.3.1.10 Target selection. With the test range output (Figure 1) connected to the video input and the test range return, with a logical zero (Item 3.1, Table I) applied to the test switch input and to the on switch inputs; with the electronic supply and readout connected as specified on Figure 5, with power supplied to the electronics supply as specified by item 2.0, Table I, and with the input signal conditions established in the item order indicated in Table V.

10.3.1.11 Minimum range inhibit 2. With the test range signal connected to the input video, with the test range return connected to the video return, with a logical zero (Item 3.1, Table I) connected to the test switch input, and with a logical zero (Item 3.1, Table I) momentarily (≥ 1 second) applied between range switch 2 and dc return, the output signal between minimum range inhibit (2) and the return shall be as specified on Figure 6.

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TABLE II. $\overline{\text{FIRE}}$ signal.

Item	Input Signal Conditions (1) (2)						Output Signal Conditions
	Reset Return	Reset	$\overline{\text{Manual Reset Switch}}$	$\overline{\text{Range Switch T}}$	$\overline{\text{Range Switch Z}}$	$\overline{\text{SRC Trigger}}$	$\overline{\text{Fire}}$
1	0	1	0	0	OC	1	1
2	1	0	1	OC	0 (3)	1	0
3	1	0	1	0	OC	0 (3)	1
4	1	0	0	0	OC	1	1
5	1	0	1	OC	0 (3)	1	0

NOTES: 1. Unless otherwise specified, referenced to dc return (item 2.8, Table I). OC = Open Current

2. Input signals: 1 = logical 1, 0 = logical 0 } Item 3.1, Table 1

3. Momentarily applied (≥ 1 second)

4. Output signals: 1 = 4.5 ± 1.0 Vdc, 0 = 0.2 ± 0.2 Vdc

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TABLE III. Output range.

Input Signal Conditions Range Counter Connector Output Lines														Output Signal Conditions			
Item	$\overline{8000}$	$\overline{4000}$	$\overline{2000}$	$\overline{1000}$	$\overline{500}$	$\overline{400}$	$\overline{200}$	$\overline{100}$	$\overline{50}$	$\overline{40}$	$\overline{20}$	$\overline{10}$	$\overline{5}$	Data 1	Data 2	Data 3	Data 4
1	O	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	--	--	--	Figure 3 e, f
2	◆	O	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	--	--	Figure 3 e, f	--
3	◆	◆	O	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	--	Figure 3 e, f	--	--
4	◆	◆	◆	O	◆	◆	◆	◆	◆	◆	◆	◆	◆	Figure 3 e, f	--	--	--
5	◆	◆	◆	◆	O	◆	◆	◆	◆	◆	◆	◆	◆	--	--	--	Figure 3 h, n
6	◆	◆	◆	◆	◆	O	◆	◆	◆	◆	◆	◆	◆	--	--	Figure 3 h, n	--
7	◆	◆	◆	◆	◆	◆	O	◆	◆	◆	◆	◆	◆	--	Figure 3 h, n	--	--
8	◆	◆	◆	◆	◆	◆	◆	O	◆	◆	◆	◆	◆	Figure 4 l, r	--	--	--
9	◆	◆	◆	◆	◆	◆	◆	◆	O	◆	◆	◆	◆	--	--	--	Figure 3 e, h
10	◆	◆	◆	◆	◆	◆	◆	◆	◆	O	◆	◆	◆	--	--	Figure 3 e, h	--
11	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	O	◆	◆	--	Figure 3 e, h	--	--
12	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	O	◆	Figure 3 e, h	--	--	--
13	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	◆	O	Figure 3 b, i	--	Figure 3 b, i	--

NOTES: 1. " " indicates a connection from the range counter connector output line through a 2 kilohm resistor (Item 1.3, Table I) to + 5 Vdc

2. "O" indicates a ground (return) connection to the specified range counter output line.

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TABLE IV. Switch and light control.

Input Signal Conditions					Output Signal Conditions			
Item	$\overline{\text{On}}$ Switch	$\overline{\text{First}}$ Switch	$\overline{\text{Second}}$ Switch	$\overline{\text{Last}}$ Switch	$\overline{\text{On}}$ Light Return	$\overline{\text{First}}$ Light Return	$\overline{\text{Second}}$ Light Return	$\overline{\text{Last}}$ Light Return
1	0	OC	OC	OC	ON	OFF	OFF	ON
2	OC	0	OC	OC	OFF	ON	OFF	OFF
3	OC	OC	OC	0	OFF	OFF	ON	OFF
4	OC	OC	OC	0	OFF	OFF	OFF	ON

NOTES: 1. Input and output signals referenced to input and output returns respectively
2. OC = open circuit
0 = logical zero, Item 3.1, Table I
ON = 0.4 + 0.4 Vdc
OFF = 13.5 + 2.0 Vdc

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TABLE V. Target selection

Input Signal Conditions (1) (2)							Output Signal Conditions	
Item	Range SW 1	Range SW 2	First SW	Second SW	Last SW	Man Reset SW	Returns	Range readout (meters ± 15)
1	0	OC	OC	OC	0 (4)	0 (4)	0	0000
2	OC	0 (4)	OC	OC	OC	NA	1	850
3	0	OC	OC	OC	OC	NA	1	850
4	OC	0 (4)	OC	NA	OC	NA	2	1850
5	0	OC	0 (4)	OC	OC	NA	2	850
6	0	OC	OC	0 (4)	OC	NA	2	1850
7	0	OC	OC	OC	OC	0 (4)	0	0000

NOTES: 1. Input signals referenced to dc input returns
2. OC = open circuit
0 = logical zero (item 3.1, Table I)
3. Output signals referenced to output return
4. Momentarily applied (≥ 1 second)
5. NA = not applicable

10.4 Test methods. All adjustments and measurements shall be made under the conditions specified in 10.3.1. Power, signals, loads, and interconnections shall be in accordance with Table I.

10.4.1 Initial turn-on. Apply a logical zero (Item 3.1, Table I) between the range switch 1 input and the dc return. Apply power (Item 2.0, Table I) and measure/verify the following:

a. The voltage waveform as specified on figure 2 shall be present between the range light return output and the return line.

b. The voltage between the last light return output and the return line shall be +0.4 ± 0.4 Vdc.

10.4.2 PFN current sensing. Adjust the PFN voltage to 1200 ± 15 V between the 1600 V coil output and the 1600 V coil return line. Measure and verify that the current between the PFN voltage sense output and the dc return line is 75 ± 2.5 microamps.

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10.4.3 PFN voltage adjustment/programming.

10.4.3.1 PFN voltage adjustment. With the 100 kilohm load connected as specified by item 1.3, Table I, measure the PFN voltage between the 1600 V coil output and the 1600 V coil return while adjusting the PFN voltage control over its range. Verify that the PFN voltage is adjustable from +1050 to +1300 Vdc.

10.4.3.2 PFN voltage programming. Adjust the PFN voltage between the +1600 V coil output and the +1600 V coil return to 1200 ± 15 V. Change the load between the thermistor and the thermistor return lines from 10 kilohms (Item 1.3, Table I) to 32.4 kilohms (Item 1.4, Table I). Measure and verify that the PFN voltage increases to 1345 ± 40 V.

10.4.4 400 V supply. Adjust the PFN voltage between the +1600 V coil output and the +1600 V coil return to 1200 ± 200 V. Measure and verify that the voltage between the +400 and +400 V return is +400 ± 40 V.

10.4.5 Fire. Adjust the PFN voltage between the +1600 V coil and the +1600 V coil return to 1200 ± 200 Vdc. Establish the input conditions of Table II in the item order indicated. Verify that the corresponding Fire signal voltage levels are as specified in Table II.

10.4.6 Range readout. Remove the range counters from the electronic supply. Establish the input signal conditions specified in Table III. Verify that the output signal conditions are as specified in Table III.

10.4.7 $\overline{\text{Manual reset}}$. Apply a logical zero (Item 3.1, Table I) between the $\overline{\text{manual reset switch}}$ and the input dc return. Measure and verify that the voltage between the $\overline{\text{manual reset}}$ and return is 0.2 ± 0.2 Vdc. Establish an open circuit at the $\overline{\text{manual reset switch}}$ input. Measure and verify that the voltage between $\overline{\text{manual reset}}$ and return is 4.5 ± 1.0 Vdc.

10.4.8 Switch and light controls. Establish the input signal conditions of Table IV in any item order. Measure and verify that the output signal conditions are as specified in Table IV.

10.4.9 Blanking. Momentarily (≥ 1 second) apply a logical zero (Item 3.1, Table I) in sequence between:

a. $\overline{\text{manual reset switch}}$ and dc return,

b. $\overline{\text{range switch 1}}$ and dc return, and

c. $\overline{\text{range switch 2}}$ and dc return.

Verify that the waveform of Figure 4 appears upon performance of step c at blanking output.

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10.4.10 Target selection. Connect the test range output (Figure 1) to the video input and the test range return to the video return. Connect the readout (Item 4.1, Table I) to the electronic supply in accordance with Figure 5. Connect power to the electronic supply (Item 2.0, Table I). Apply a logical zero (Item 3.1, Table I) to the test switch inputs. Establish the input signal conditions as specified in Table V in the item order indicated. Verify that the corresponding output signal conditions are as specified in Table V.

10.4.11 Minimum range inhibit 2. Connect the test range output (Figure I) to the video input and the test range return to the video return. Apply a logical zero (Item 3.1, Table I) to the test switch input. Momentarily (≥ 1 second) apply a logical zero (Item 3.1, Table I) in sequence between:

a. range switch I and dc return, and

b. range switch 2 and dc return.

Verify that the waveform of Figure 6 appears upon performance of step b at the minimum range inhibit (2) output.

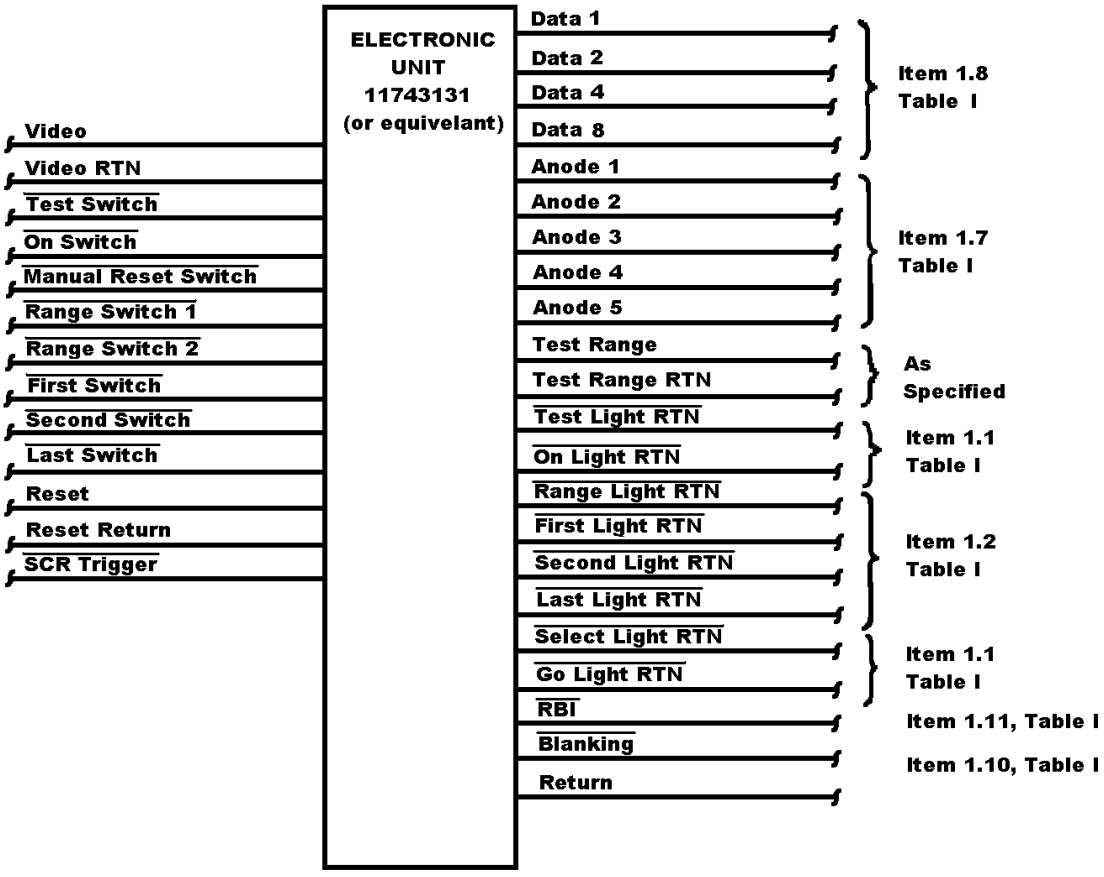


FIGURE 1. Electronic supply power, signals, and loads (sheet 1 of 2).

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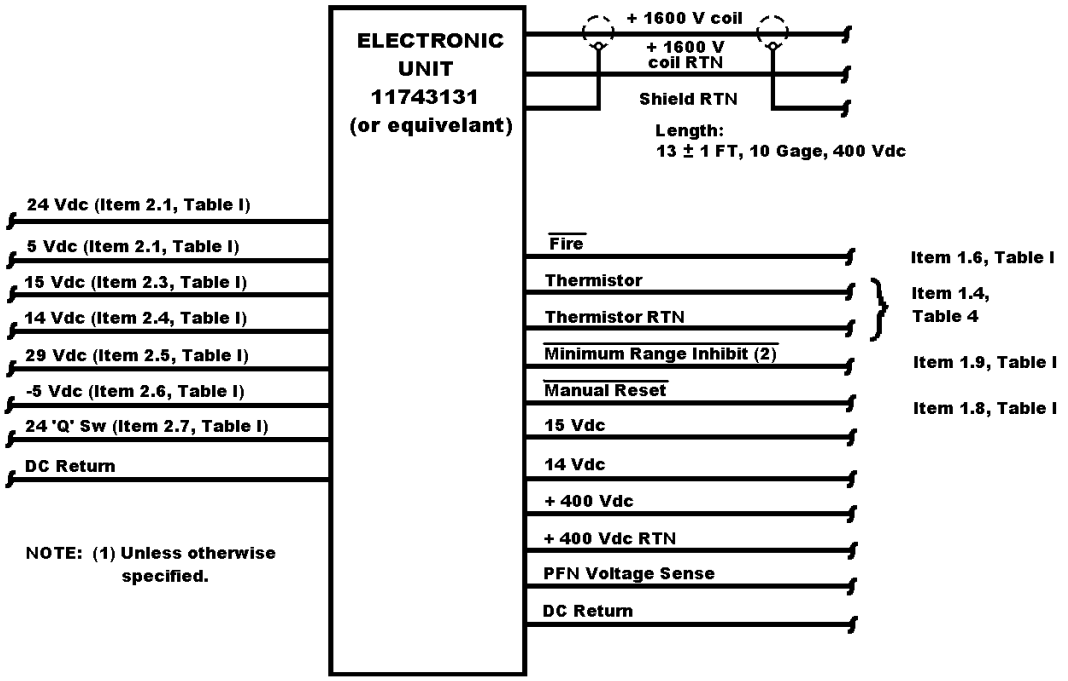


FIGURE 1. Electronic supply power, signals, and loads (sheet 2 of 2).

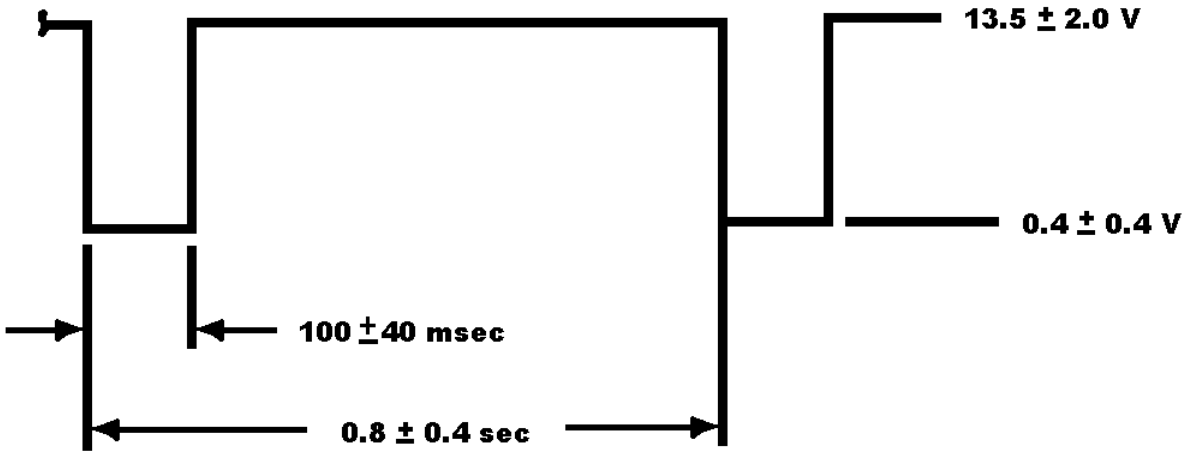
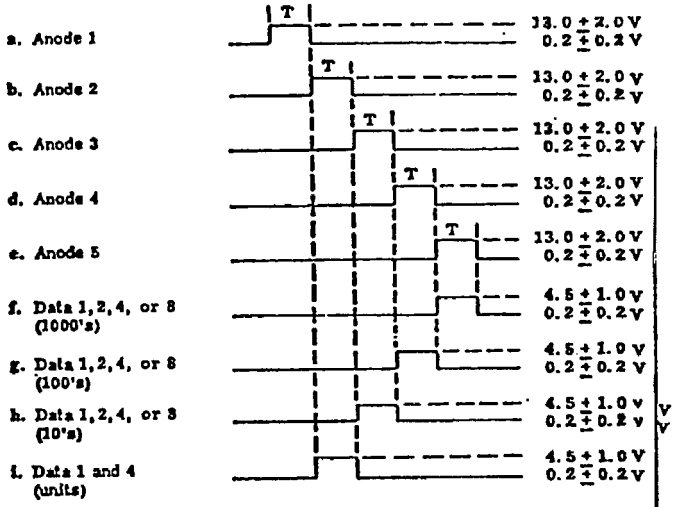


FIGURE 2. Range light return waveform.

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NOTES: 1. T = 3.2 ± 0.9 msec
2. Voltages referenced to output return line.

FIGURE 3. Anode 1 through 5 and data 1, 2, 4, and 8 signals.

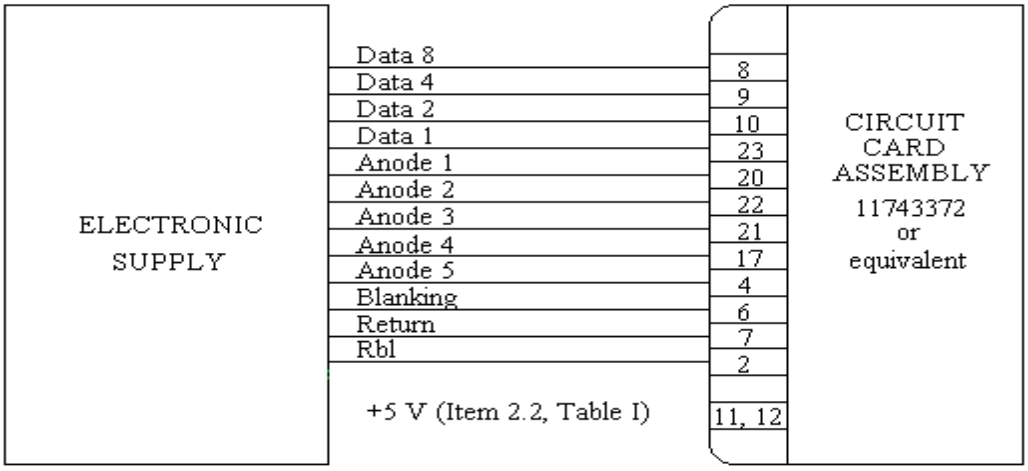


FIGURE 5. Electronic Supply and Readout connections.

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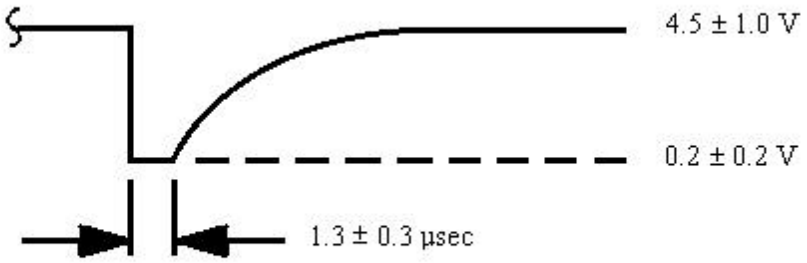


FIGURE 6. Minimum range inhibit (2).

”

Add Appendix II following Appendix I as shown:

“APPENDIX II
MIRROR CONTROL UNIT REQUIREMENTS

20.1 SCOPE

20.1.1 This specification establishes the performance requirements for the Mirror Control Unit Special Test Equipment.

20.2 APPLICABLE DOCUMENTS

DRAWINGS

Military

F11732377	Mirror Drive Electronics

20.3 REQUIREMENTS

20.3.1 Performances. The Mirror Control Unit Special Test Equipment shall meet the requirements of this specification under the following conditions:

a. Temperature	73 ± 18°F
b. Pressure	28.5 +2.0 or –4.5 inches of mercury
c. Relative humidity	50 ± 30 percent
d. Power, Signals, Loads and Interconnections	See Table I

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20.3.1.1 Motor drive gain. With the load, power, and signals of Table I applied as specified therein and the input signals of table II adjusted as specified therein for each input condition, the corresponding differential outputs at the load shall be as specified in Table II.

20.3.1.2 Null condition. With D DOT L adjusted to 0.750 ± 0.001 Vrms and DL FDBK adjusted to 0.026 ± 0.001 Vrms in phase with D DOT L, the output at the load shall be less than 0.300 Vrms.

Table I. Load, power, and signals

Item	Conditions	Characteristics	Connections
1	Load	See Figure I	As specified on Figure 2
2	Power Sources		
2.1	+24 Vdc	+24.0 ± 1.0 Vdc Ripple: ≤ 200 mV p-p	
2.2	+15 Vdc	+15.0 ± 0.7 Vdc Ripple: ≤ 25 mV p-p	
2.3	-15 Vdc	-15.0 ± 0.7 Vdc Ripple: ≤ 25 mV p-p	
3	Signals		
3.1	AC Ref 1	5.000 ± 0.002 Vrms 400 ± 4 Hz	
3.2	AC Ref 2	5.000 ± 0.002 Vrms 400 ± 4 Hz 180° out of phase with AC Ref 1	
3.3	DL FDBK	Sinusoidal waveform with variable amplitude	
3.4	Position Command		
3.5	D DOT L		

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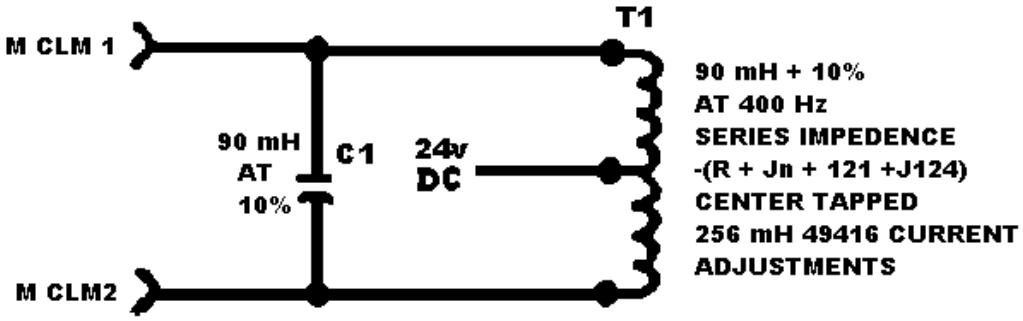


FIGURE 1. Load circuit

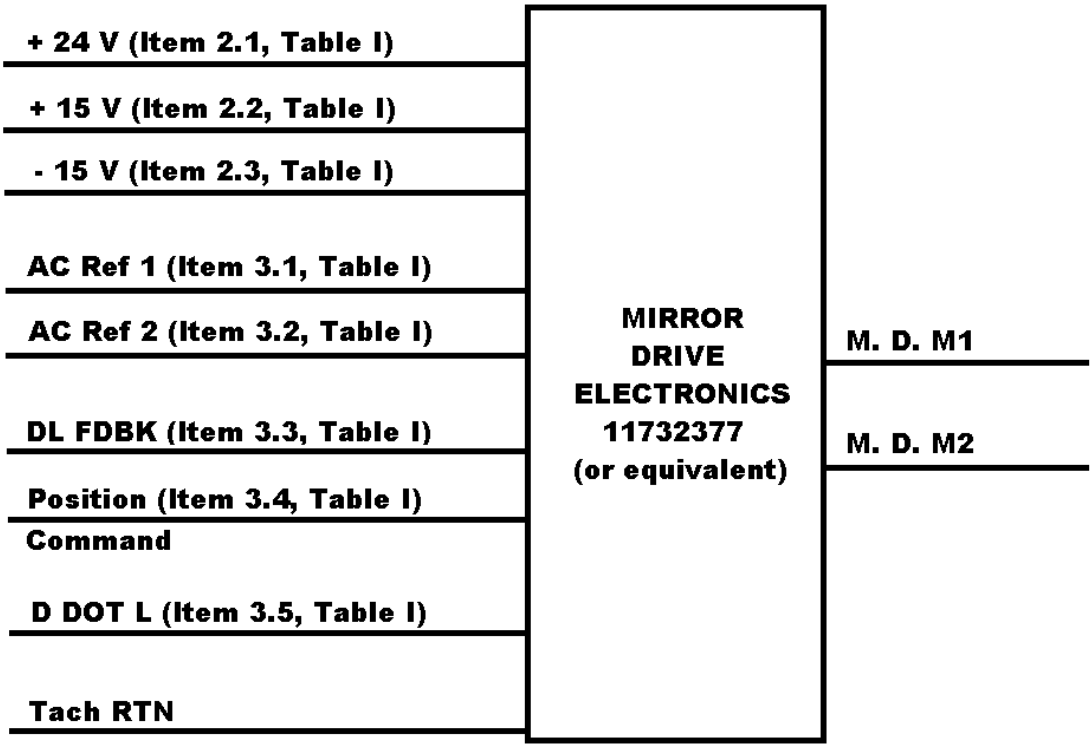


FIGURE 2. Mirror control unit, input output interface.

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Table II. Motor drive out.

Item	Input Signal Condition			Output
	DL FDBK	Position Command	D DOT L	M.D. M1 – M.D. M2
1	0.053 ± 0.001	0.000 ± 0.001	0.000 ± 0.001	15.00 ± 0.50
2	0.000 ± 0.001	0.053 ± 0.001	0.000 ± 0.001	15.00 ± 0.50
3	0.000 ± 0.001	0.000 ± 0.001	1.52 ± 0.10	15.00 ± 0.50

Notes:
1. Input and output signals: sinusoidal, Vrms
2. Grd = Ground

20.4 Test methods. All measurements shall be made under the conditions specified in 20.3.1. Unless otherwise specified, power, signals, and load shall be in accordance with Table I.

20.4.1 Motor drive gain. Adjust the input signals as specified in Table II, and measure the output voltage between M.D. M1 and M.D. M2 for each condition to determine compliance with 20.3.1.1.

20.4.2 Null condition. Adjust the input signals as specified in 20.3.1.2, and measure the output voltage between M.D. M1 and M.D. M2 to determine compliance with 20.3.1.2.

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